Step 1: Verify Java connectivity and version

Open Terminal.

```
None
java -version
```

Check Java installation:

```
None
echo $JAVA_HOME
```

For your Hive environment, verify that the configured Java version is compatible with Hive 4.1.0.

Step 2: Verify Hive installation

```
None
hive --version
```

Expected: Hive 4.1.0.

Check the Hive environment variable:

```
None
echo $HIVE_HOME
```

If it is not set, check your Homebrew installation:

```
None
brew --prefix hive
```

Step 3: Verify Hadoop connectivity

Check Hadoop:

```
None
hadoop version
```

Check HDFS:

```
None
hdfs dfs -ls /
```

If HDFS is not running, check:

```
None
jps
```

Look for:

- NameNode
- DataNode
- SecondaryNameNode

If you are using local filesystem mode rather than HDFS, this HDFS check may not be required.

Step 4: Check Hive Metastore

Hive Metastore stores metadata about databases, tables, columns and partitions.

Check whether the metastore process is running:

```
None
jps
```

Look for a process named `HiveMetaStore`.

Check the metastore port:

```
None
lsof -i :9083
```

Default Hive Metastore Thrift port: 9083.

If it is not running, start it in a separate Terminal window:

```
None
hive --service metastore
```

Keep this terminal open.

Step 5: Verify HiveServer2

HiveServer2 is the service that Beeline connects to.

Check whether port 10000 is active:

```
None
lsof -i :10000
```

Or:

```
None
nc -zv localhost 10000
```

Expected successful output:

```
None
Connection to localhost port 10000 succeeded!
```

Connected
Port 10000 is accessible.

If the port is not listening, start HiveServer2 in another Terminal window:

```
None
hive --service hiveserver2
```

Wait until the service starts. Keep this terminal open.

Step 6: Verify connectivity before Beeline login

Execute:

```
None
nc -zv localhost 10000
```

You can also check both Hive ports:

```
None
nc -zv localhost 9083
nc -zv localhost 10000
```

Port	Component	Purpose
9083	Hive Metastore	Metadata connectivity
10000	HiveServer2	Beeline JDBC connectivity

Step 7: Connect using Beeline

Now execute:

```
None
beeline -u jdbc:hive2://localhost:10000/default
```

If authentication is required:

```
None
beeline -u jdbc:hive2://localhost:10000/default \
-n hive
```

Expected:

```
None
Connected to: Apache Hive
Driver: Hive JDBC
Beeline version 4.1.0
```

Step 8: Execute a test query

Once connected:

```
None
SHOW DATABASES;
```

Then:

```
None
CREATE DATABASE IF NOT EXISTS training_db;
```

```
None
USE training_db;
```

```
None
CREATE TABLE test_table (
```

```
    id INT,
    name STRING
);
```

Verify:

```
None
SHOW TABLES;
```

Step 9: Execute your SQL file

Exit Beeline:

```
None
!quit
```

Execute your file:

```
None
beeline -u jdbc:hive2://localhost:10000/default \
-n hive \
-f employee.sql
```